GENERAL PHARMACOLOGY

MBS230-P&T

UNIT 1.2.0

John Chitoti
Lecturer, Pharmacology
School of Medicine
Copperbelt University

Pharmacodynamics

LS 1.2.0 MECHANISMS OF DRUG ACTION

LEARNING OBJECTIVES

Students must be able to:

- Understand molecular targets for drug action
- Explain mechanisms by which drugs produce their effects
- Explain the differences between drug specificity and drug selectivity

MECHANISMS OF DRUG ACTION .... CONT'D

At the end of the Lecture: students should be able to:

- List the molecular targets for drug action
- Describe the various mechanisms by which drugs produce their effects
- Differentiate between drug specificity and drug selectivity

MECHANISMS OF DRUG ACTION

INTRODUCTION:

The **mechanism of action** of a drug are the processes involved in the interaction between a drug and specific cellular macromolecules at the site of action to produce a biological effect

Mechanisms of drug action can be viewed from two perspectives, namely, the site of action and the nature of the interaction between the drug and its molecular targets at the site of action

The molecular interactions at the drug target site modifies cellular functions resulting in a biological effect

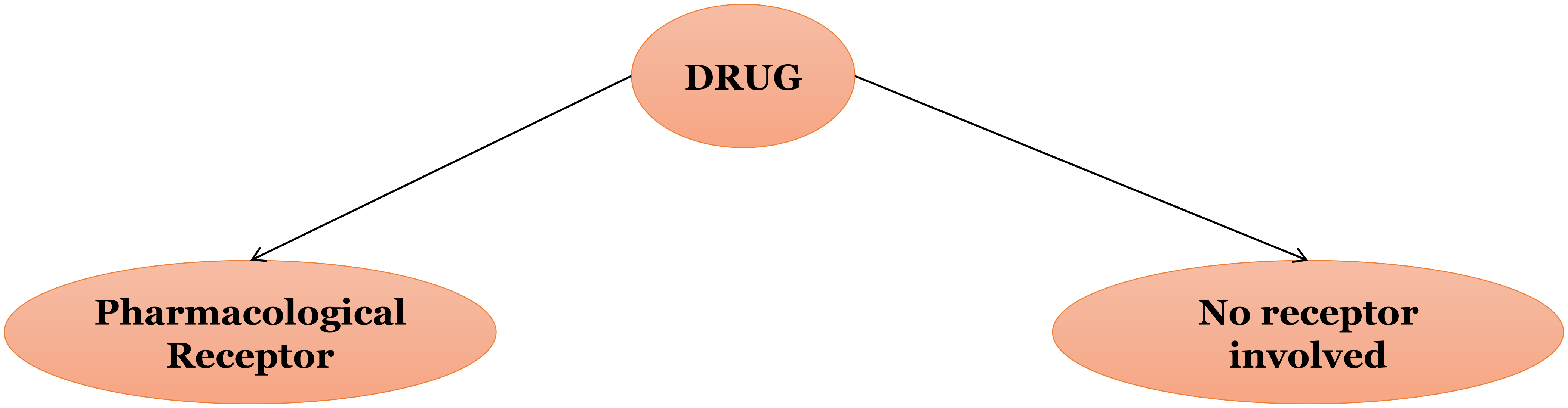
Mechanism of Drug Action

Pharmacodynamics

- Pharmacodynamics is concerned with actions, interactions and the mechanism (mode) of action of drugs. some drug receptor interaction are highly specific

➢ Pharmacological receptors are involved

➢ some are non specific (receptors not involved)



MECHANISMS OF DRUG ACTION …. CONT'D

INTRODUCTION …. CONT'D:

The resulting effects from drug action is either due to stimulation or depression of normal physiological functions. Stimulation increases the rate of activity while depression reduces the rate of activity.

Drug actions result in an increase, a decrease or replacement of enzymes, hormones or other chemical mediators of cellular functions.

Most drugs act on target proteins i.e. receptors, enzymes, carriers, membrane bound transport proteins and pumps, ion channels etc.

SITES OF DRUG ACTION: RECEPTORS

- Most drugs produce their effects through interacting with receptors
- A receptor is the specific macromolecular component (sensing element) of the cell with which endogenous chemical mediators and drugs interact to produce its biological effects
- E.g. Chemical mediators; Hormones, neurotransmitters, autocoids, histamine and seretonin.
- A drug may act either my mimicking endogenous mediators and activating the receptor (agonists) or bind on the receptor and prevent receptor activation by the endogenous mediators (antagonists)

SITES OF DRUG ACTION: ENZYMES

Enzymes are protein catalysts that increase the rate of specific chemical reactions without undergoing any net change themselves during the reaction

Some drugs act by modifying normal biochemical reactions through enzyme inhibition

Enzyme inhibition may be:

- Competitive and either reversibly binding(e.g. neostigmine acting on acetylcholinestrase) or non-competitive and irreversibly binding (asprin acting on cyclo-oxygenase)

SITES OF DRUG ACTION: ENZYMES …. CONT'D

Examples of drugs that act by enzyme inhibition:

- Mono-amino oxidase inhibition by phenelzine
- Cholinesterase inhibition by pyridostigmine
- Xanthine oxidase inhibition by allopurinol
- Cyclo-oxygenase inhibition by aspirin
- Angiotensin converting enzyme inhibition by captopril

ION CHANNELS

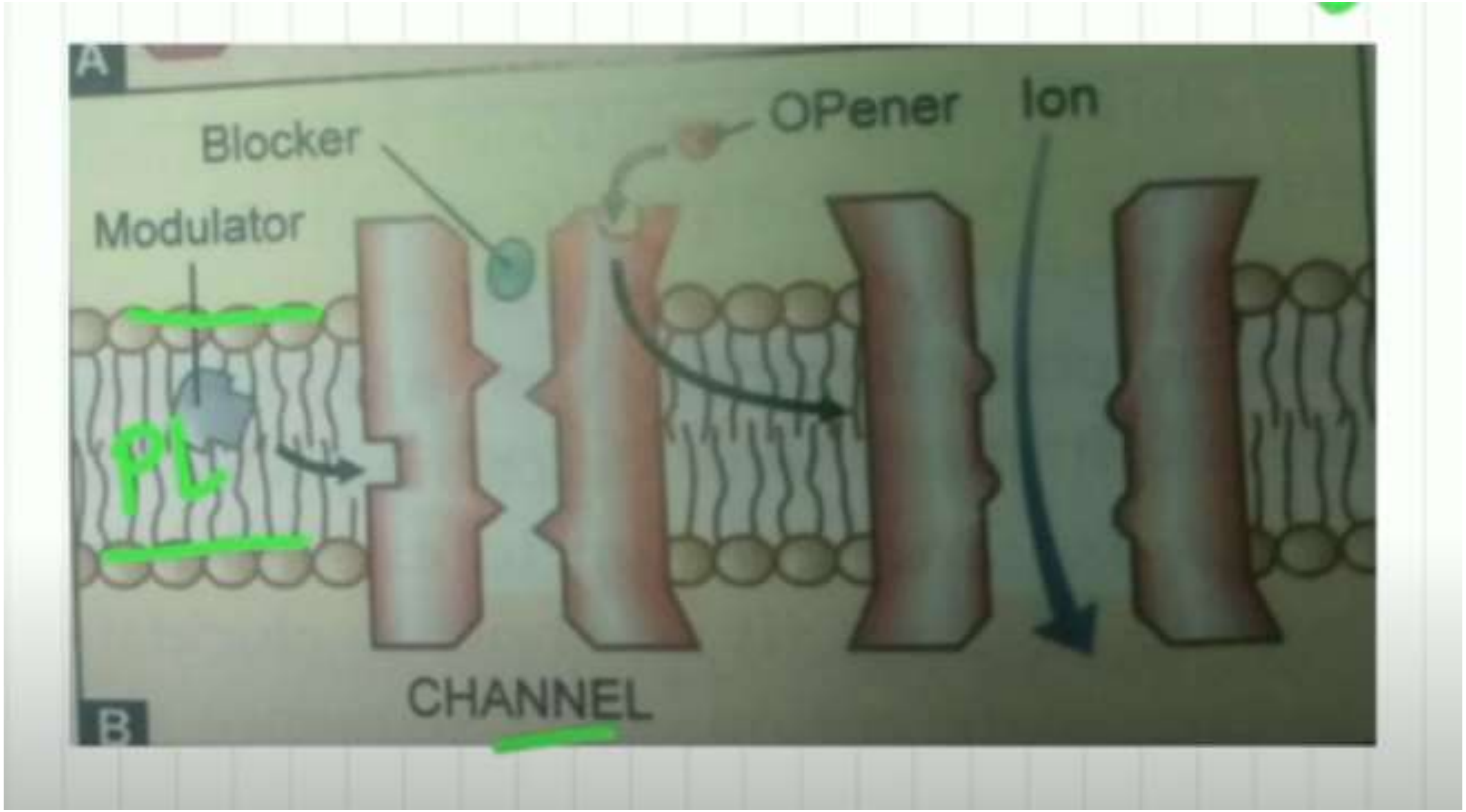
- Ion channels (ligand-gated ion channel) are pores located in the cell membranes that allow selective transfer of ions in and out of the cell to regulate cellular activities
- Opening or closing of these channels (gating) occurs as a result of the channel proteins undergoing a change in shape after an agonist binds to the ion channel.
- Some drugs produce their effects through interactions with ion channels
- Some drugs modify ion channel function directly by binding to a part of the channel protein while others do so indirectly through intermediate molecules such as G-proteins

ION CHANNELS …. CONT'D

Examples of drugs that act by interactions with ion channels:

- Local anaesthetics that act by blocking sodium channels on neuronal membranes (e.g. lignocaine)
- Drugs that block calcium channels in vascular smooth muscle cell membranes resulting in vasodilatation (e.g. amlodipine)
- Drugs that open potassium channels on vascular smooth muscle cell membranes resulting in vasodilatation (e.g. diazoxide and minoxidil)

ION CHANNELS



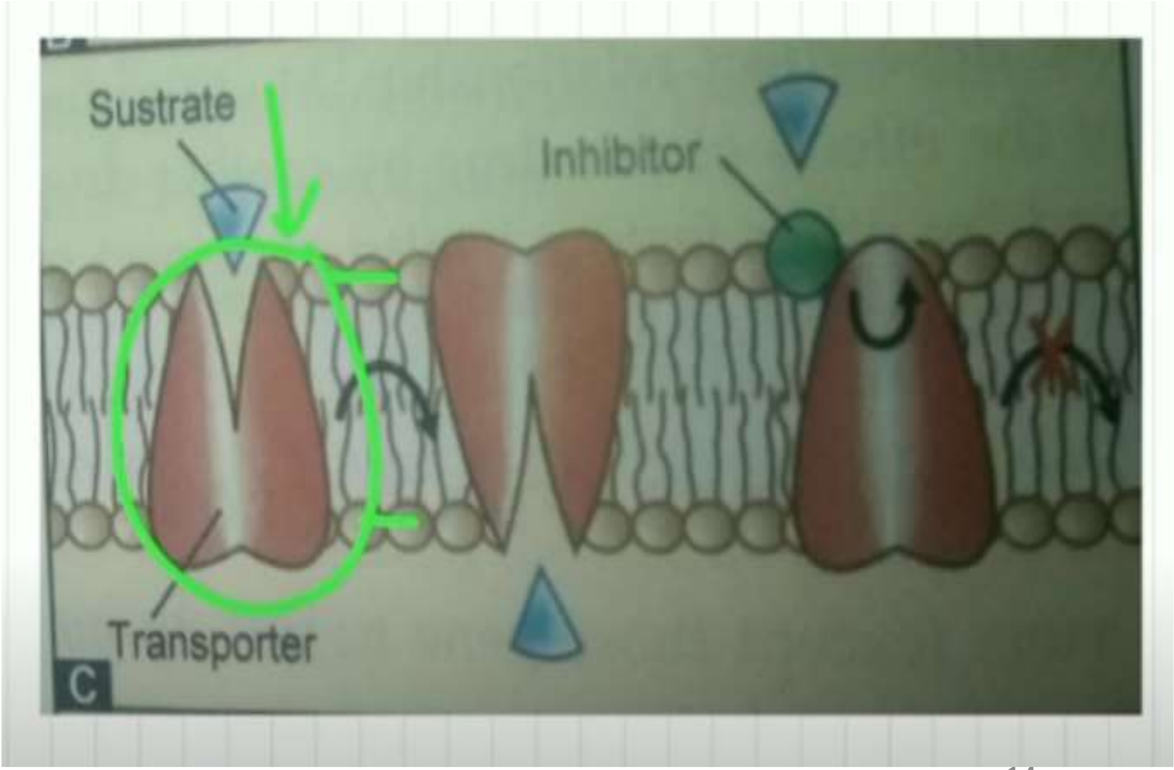
CARRIER PROTEINS

- Transfer of some ions and molecules across cell membranes is facilitated by carrier proteins located in the cell membrane
- Some drugs act by interactions with carrier proteins thereby interfering with the passage of substances across cell membranes
- Glucose & amino acid transporter
- neurotransmitter precursors

e.g. (Choline)

- neurotransmitter uptake

e.g. (5-HT, Glutamate uptake)



CARRIER PROTEINS …. CONT'D

There are two types of carrier proteins:

1. Energy-dependent carriers, termed pumps, e.g. Na^+/K^+ ATPase
2. Energy-independent carriers
   - Transporters: move one type of ion/molecule in one direction
   - Symporters: move two or more ions/molecules
   - Antiporters: exchange one or more ions/molecules for one or more other ions/molecules

CARRIER PROTEINS …. CONT'D

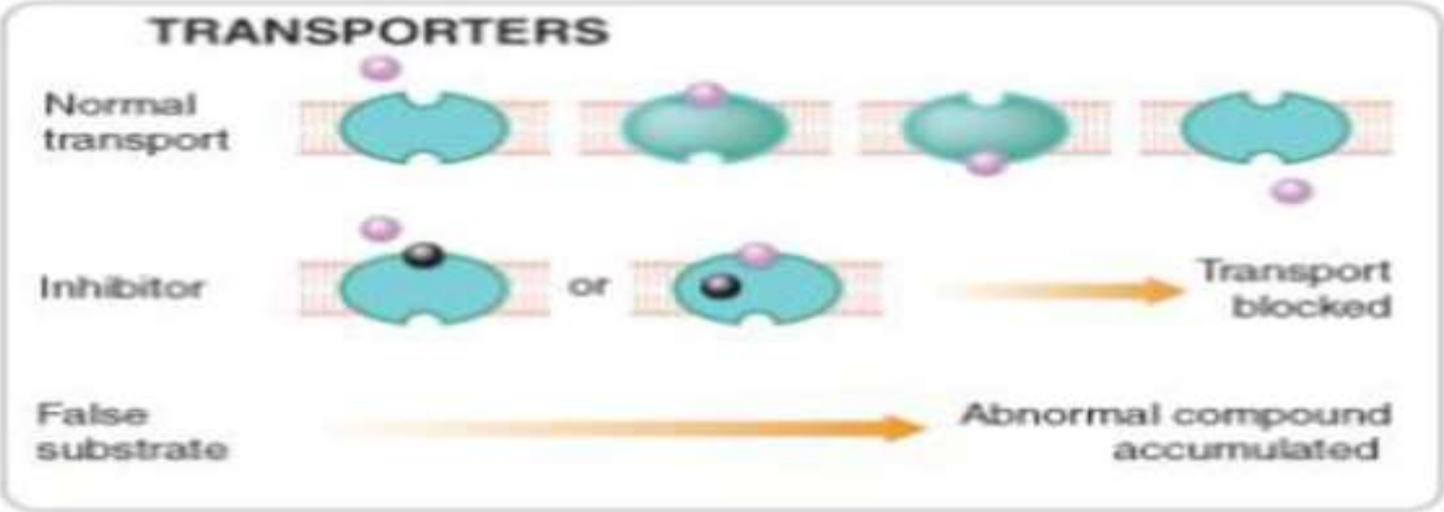
Examples of drugs that act by interacting with carrier proteins

- Inhibition of membrane bound Na^+/K^+ ATPase by cardiac glycosides
- Inhibition of membrane bound H^+/K^+ ATPase by omeprazole
- Inhibition of the renal tubule weak acid carrier by probenecid
- Inhibition of the loop of Henle $Na^+/K^+/Cl^-$ co-transporter by loop diuretics

•The amine transporters belong to a well-defined structural family, distinct from the corresponding receptors.

The carrier proteins embody a recognition site that makes them specific for a particular permeating species.

These recognition sites can also be targets for drugs whose effect is to block the transport system. e.g. TCA, Cocaine, Omeprazole, Cardiac Glycosides



NON-SPECIFIC INTERACTIONS

Drugs that act by means of their physicochemical properties are considered to have non-specific mechanisms of action

Examples of drug action resulting from physical properties

- Action by osmotic properties e.g. laxatives such as magnesium sulphate and diuretics such as mannitol
- Action by adsorption e.g. activated charcoal in posisoning
- Action by radio-activity e.g. radio-active iodine (I^{131})

NON-SPECIFIC INTERACTIONS …. CONT'D

Examples of drug action resulting from physical properties from chemical interactions

- Chelating agents bind heavy metals rendering them inactive and more readily excreted (e.g. EDTA in heavy metal poisoning) [EDTA = Ethylene-diamine-tetra-acetic acid]

- Neutralization of gastric acid by antacids

CHEMOTHERAPEUTIC AGENTS

- Chemotherapy is the drug treatment for the diseases caused by bacteria and the other pathologic microorganisms, parasites, and tumor cells; without damaging host tissues
- Chemotherapeutic agents act by killing or weakening foreign organisms such as fungi, bacteria, worms, viruses or malignant cells
- The main principle of action is selective toxicity, i.e. the drug must be more toxic to the parasite than to the host
- Selective toxicity is achieved by exploiting basic biochemical and physical differences between the infectious organism and the host cells

CHEMOTHERAPEUTIC AGENTS …. CONT'D

Anti-microbial agents act by:

- Alteration of metabolic processes unique to micro-organisms (e.g. penicillins inhibit the bacterial cell wall)
- Having enormous quantitative differences in affecting a process common to both humans and microbes (e.g. inhibition of folic acid metabolism by trimethoprim)

Cytotoxic drugs act by interfering with cell growth and division at different stages of the cell cycle

MECHANISMS OF ACTION OF CHEMOTHERAPEUTIC AGENTS

- Inhibition of cell wall synthesis
- Inhibition of functions of cellular membrane
- Inhibition of nucleic acid (DNA and RNA) synthesis
- Inhibition of protein synthesis
- Inhibition of folic acid synthesis and utilization
- Disruption of DNA structure and function
- Inhibition of microtubule function

SPECIFICITY AND SELECTIVITY OF DRUG ACTIONS

- If a drug has one effect, and only one effect on all biological systems it possesses the property of **specificity.** In this case, the drug acts on only one molecular target.

- A drug is regarded as non-specific when it acts on more than one molecular target

- The vast majority of drugs are **selective** rather than specific. This implies that they can act on more than one molecular target once they reach an appropriately high concentration.

- Selectivity of a drug depends on the concentration it achieves in various tissues

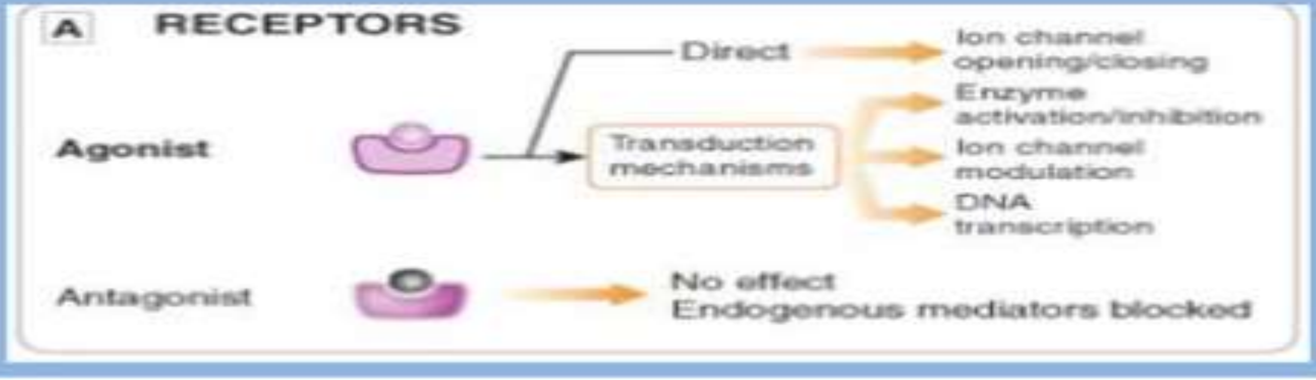
In general, the drug-receptor interaction is characterized first by binding of drug to receptor and second by generation of a response in a biological system. The first function is governed by the chemical property of affinity, ruled by the chemical forces that cause the drug to associate reversibly with the receptor.

Affinity

It describes the ability of a drug to form and subsequently maintain a complex with a receptor site.

Intrinsic Activity

It describes the ability of a drug to evoke a pharmacologic response on combining with a receptor. E.g. Acetylcholine acts as a cholinergic agonist.



LEARNING OUTCOMES

Students must be able to comprehend:

- The molecular targets for drug action
- Mechanisms by which drugs produce their effects
- The difference between drug specificity and drug selectivity